

11(a)	substance containing radioactive nuclei that is introduced into the body <i>or</i> substance containing radioactive nuclei that is absorbed by the tissue being studied	B1
11(b)(i)	a particle interacting with its antiparticle so that mass is converted into energy	B1
11(b)(ii)	electron(s) and positron(s)	B1
11(c)(i)	$E = 2mc^2$ $= 2 \times 9.11 \times 10^{-31} \times 3.00 \times 10^8$ $= 1.64 \times 10^{-13} \text{ J}$	A1

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Question	Answer	Marks
11(c)(ii)	$\lambda = \frac{2hc}{E}$ $= \frac{2 \times 6.63 \times 10^{-34} \times 3.00 \times 10^8}{1.64 \times 10^{-13}}$	C1
	$= 2.43 \times 10^{-12} \text{ m}$	A1
11(d)	Any 3 from: - the two gamma photons travel in opposite directions - gamma photons detected (outside body / by detectors) - gamma photons arrive (at detector) at different times - determine location of production (of gamma) - image of tracer concentration in tissue produced	B3

Question	Answer	Marks
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